Duc-Tuan Truong

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Overview

Final-year Ph.D. student focused on speech AI, with research in deepfake speech detection, speaker recognition, and speaker age estimation. Experienced in developing new training pipelines and model architectures, leveraging state-of-the-art self-supervised speech foundation models and speech LLMs to improve downstream tasks.

Education

- **Nanyang Technological University** — Singapore
  Doctor of Philosophy (Ph.D.) in Computer Science; GPA: 4.83/5.0 — *Jan 2023 - Jan 2027 (expected)*
- **Nanyang Technological University** — Singapore
  Master of Science in Artificial Intelligence; GPA: 4.55/5.0 — *Jan 2022 - Jan 2023*
- **Ho Chi Minh University of Technology (HCMUT) - Vietnam National University** — HCMC, Vietnam
  Bachelor of Engineering in Computer Science; GPA: 8.57/10.0 — *Aug 2017 - Aug 2021*
  One of 40 selected students among 300 Computer Science students for the Talented Engineer Program.

Work Experience

- **American Express Innovation Labs** — Singapore
  Incoming Data Science Intern — *May 2026 - Aug 2026*
- **Hong Kong Polytechnic University** — Hong Kong
  Research Student Attachment Programme — *Feb 2025 - Jun 2025*
  - Designed and evaluated novel training techniques for deepfake speech detection [[5]].
  - Implemented a proof-of-concept explainable Deepfake Speech Detection system using SpeechLLM Qwen2.5 Omni model.
- **Zalo AI** — HCMC, Vietnam
  Associate AI Engineer (Full-time) — *Aug 2021 - Jan 2022*
  - Contributed to *KiKi* - the first Vietnamese voice assistant deployed on smart speakers, cars, and smartphones.
  - Parsed ASR outputs into intents and implemented intent-ranking logic for voice queries.
  - Built a streaming pipeline to update NER entities by crawling and processing data from multiple websites.
  - Analyzed & visualized ∼150,000 daily user logs to diagnose failure cases and prioritize fixes.
- **Innotech** — HCMC, Vietnam
  AI Engineer (Full-time) — *Nov 2020 - Jul 2021*
  - Built centralized log management and deployed AI models for streaming anomaly detection on servers using log messages.
  - Adapted reinforcement learning (Deep Q-learning, AlphaZero) to solve FreeCell.

Publications (⋆ denotes co-first author) (Total citations: 150+, H-index: 6)

[1] **D.-T. Truong**, R. Tao, H.-T Luong, T. Nguyen, K.A. Lee, C.E. Siong, *Temporal-Channel Modeling in Multi-head Self-Attention for Synthetic Speech Detection*, **ISCA INTERSPEECH 2024**, *GitHub stars: 55*.

[2] **D.-T. Truong**, R. Tao, J.Q Yip, K.A. Lee, C.E. Siong, *Emphasized Non-Target Speaker Knowledge in Knowledge Distillation for Automatic Speaker Verification*, **IEEE ICASSP 2024**, *GitHub stars: 16*.

[3] T. Liu, **D.-T. Truong**, R.K. Das, K.A. Lee, H .Li, *Nes2Net: A Lightweight Nested Architecture for Foundation Model Driven Speech Anti-spoofing*, **IEEE Transactions on Information Forensics and Security 2025**, *GitHub stars: 103*.

[4] R. Tao, Z. Shi, Y. Jiang, **D.-T. Truong**, C.E. Siong, M. Alioto, H. Li, *Multi-Stage Face-Voice Association Learning with Keynote Speaker Diarization*, **ACM Multimedia 2024**, *GitHub stars: 22*.

[5] **D.-T. Truong**, T. Liu, J. Li, R. Tao, K.A. Lee, C.E. Siong, *Addressing Gradient Misalignment in Data-Augmented Training for Robust Speech Deepfake Detection*, **IEEE ICASSP 2026**, *GitHub stars: 3*.

[6] T.D Phuong⋆, **D.-T. Truong**⋆, L.V. Hoang, T.T.T. Nguyen, *Fine-grained Frame Modeling In Multi-head Self-attention For Speech Deepfake Detection*, **IEEE ICASSP 2026**.

[7] C.Y Kwok⋆, **D.-T. Truong**⋆, R.K. Das, K.A. Lee, H. Li, *Robust Audio Deepfake Detection Using Ensemble Confidence Calibration*, **IEEE ICASSP 2025**.

[8] **D.-T. Truong**, T.T. Anh, C.E. Siong, *Exploring Speaker Age Estimation on Different Self-Supervised Learning Models*, IEEE **APSIPA ASC 2022**, *GitHub stars: 14*.

[9] J.Q Yip, **D.-T. Truong**, D. Ng, C. Zhang, Y. Ma, T.H. Nguyen, C. Ni, S. Zhao, C.E. Siong, B. Ma *ACA-Net: Towards Lightweight Speaker Verification using Asymmetric Cross Attention*, **ISCA INTERSPEECH 2023**.

[10] J. Li, K.A. Lee, **D.-T. Truong**, T. Liu, R. Tao, M.W. Mak, *Xi+: Uncertainty Supervision for Robust Speaker Embedding*, **IEEE ICASSP 2026**.

[11] **D.-T. Truong**, T. Liu, R. Tao, J. Li, K.A. Lee, C.E. Siong, *QAMO: Quality-aware Multi-centroid One-class Learning For Speech Deepfake Detection*, **preprint 2026**, *GitHub stars: 6*.

[12] T. Gupta, **D.-T. Truong**, T.T. Anh, C.E. Siong, *Estimation of speaker age and height from speech signal using bi-encoder transformer mixture model*, **ISCA INTERSPEECH 2022**, *GitHub stars: 14*.

[13] H.-T. Luong, **D.-T. Truong**, K.A. Lee, C.E. Siong, *Room impulse responses help attackers to evade deep fake detection*, **IEEE SLT 2024**.

[14] **D.-T. Truong**, Y. Wang, K.A. Lee, M. Li, H.Nishizaki, C.E. Siong, *A Study of Guided Masking Data Augmentation for Deepfake Speech Detection*, **ASVspoof Workshop 2024**.

Projects

- **Deepfake Speech Detection (Research Project funded by AI Singapore)**
  Developed deepfake speech detection methods that enhance both architecture and optimization. Designed temporal–channel modeling for multi-head self-attention [[1]] and addressed gradient misalignment introduced by data augmentation [[5]], improving robustness and cross-condition generalization for synthetic speech detection.
- **Homomorphic Encryption Speaker Verification (Research Project funded by OCBC Bank)**
  Implemented an end-to-end speaker verification pipeline, training SOTA models and enabling privacy-preserving inference under fully homomorphic encryption.
- **Interpretability of Speech Signal under Adverse Conditions (Research Project funded by Temasek Lab)**
  Developed a novel knowledge distillation-based training method that transfers knowledge from large robust speaker verification models to compact, deployment-friendly models [[2]].
- **Identifying Speaker Characteristics through Audio Profiling (Research Project funded by Temasek Lab)**
  Developed a gender-aware mixture-of-experts Transformer model that integrates self-supervised speech representations to accurately estimate speaker characteristics like age and height directly from speech signals [[8]], [[12]].
- **Multi-camera multi-people tracking (Undergraduate Thesis)**
  Implemented system a tracking people across multiple non-overlapping cameras using DeepSORT for tracking and ABD-Net model for feature extraction.

Honors and Awards

- 1st place winner in FAME Challenge, ACM-Multimedia, 2024.
- ISCA INTERSPEECH Travel Grant, 2024.
- NTU Research Scholarship (full-ride scholarship awarded to graduate students).
- KSYS-CUBE Scholarship 2021 (Awarded to 40 excellent HCMUT students annually).
- Acecook Happy Scholarship 2021 (Awarded to 200 outstanding students among 17 universities across Vietnam).
- TRUaire Scholarship 2019 & 2020 (Awarded to 40 selected HCMUT students annually).

Skills Summary

- **Languages:** Python, Java, C++, Bash, SQL, LaTeX
- **Frameworks:** HuggingFace, ms-swift, fairseq, PyTorch, Kafka, Faust, MLflow, Vert.x, Django
- **Tools:** Docker, Elasticstack, PostgreSQL, MySQL, Vim, Git

Languages

- **English:** Fluent (IELTS Academic: 7.5).
- **Vietnamese:** Native.

Academic Service

Reviewer: ICASSP (2026), INTERSPEECH (2026), SLT (2026).

References

- Prof. Chng Eng Siong - NTU (Singapore) - ASESChng@ntu.edu.sg - Main supervisor
- Prof. Lee Kong Aik - PolyU (Hong Kong) - kongaik.lee@polyu.edu.hk - Co-supervisor
- Dr. Tao Ruijie - Zoom (Singapore) - ruijie.tao@u.nus.edu - Mentor
- Dr. Liu Tianchi - Tencent (Singapore) - tianchi_liu@u.nus.edu - Mentor